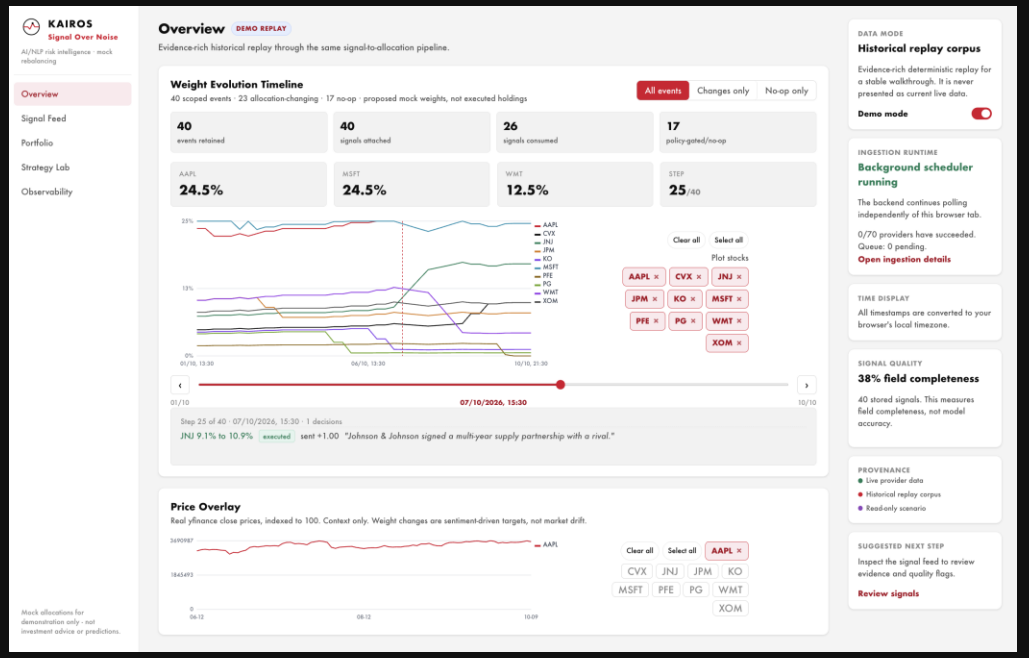
KAIROS

Signal Over Noise

FROM FINANCIAL TEXT TO AUDITABLE DECISIONS

- Company-specific, evidence-backed risk signals
- Policy-gated simulated portfolio rebalancing



THE ANALYST PROBLEM

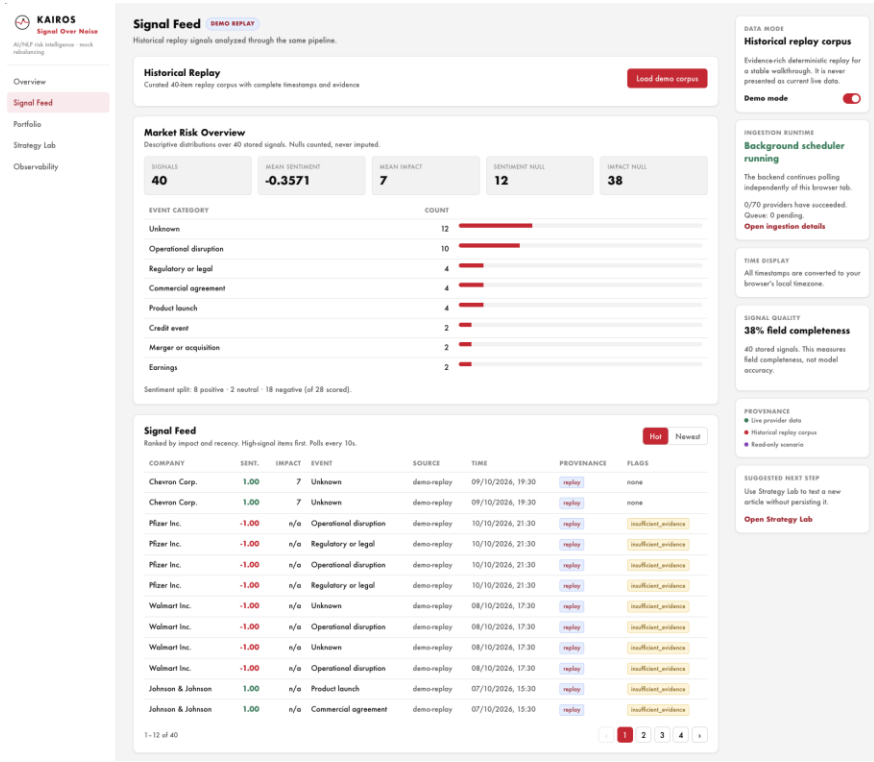
A headline is not yet a portfolio decision

ILLUSTRATIVE FINANCIAL EVENT

“Apple cuts revenue guidance.”

What must an analyst establish?

- Which holding is affected?
- What happened, and in which direction?
- What source evidence supports magnitude?
- Does policy permit a mock weight change?



KAIROS connects interpretation to a reviewable decision—not an automatic trade.

SYSTEM DESIGN

One traceable path: source → decision

01	SOURCES	GDELT · RSS · StockTwits · Reddit RSS · SEC EDGAR · yfinance
02	INGEST	Background polling · normalize · deduplicate · bounded queue
03	INTERPRET	Resolve company · sentiment · event · exact evidence
04	CONTROL	Deterministic impact rubric · signal validation · Module A gates
05	EXPOSE & AUDIT	SQLite signal/decision stores · FastAPI REST/SSE · React

Deployment: FastAPI + React/Nginx in Docker Compose. Scheduler runs independently of the browser.

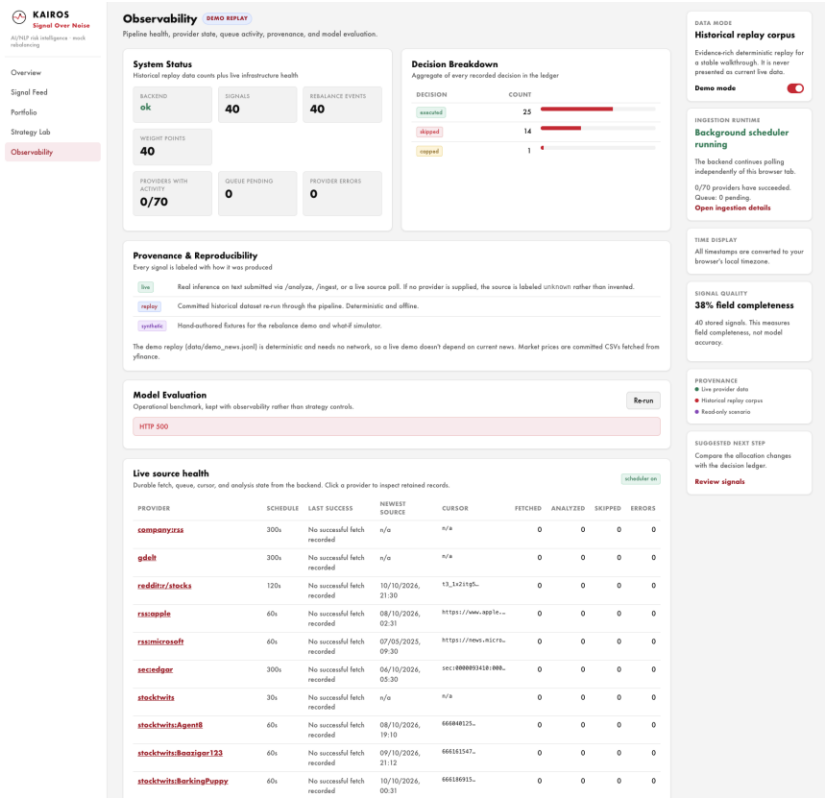
ENGINEERING CHALLENGES & SOLUTIONS

What can go wrong—and how KAIROS responds

CHALLENGE 01 | AMBIGUOUS FINANCIAL TEXT
Company aliases and cashtags resolve holdings; hybrid NLP identifies sentiment and events.
Exact excerpts retained from source text

CHALLENGE 02 | UNSUPPORTED IMPACT
Deterministic factor-based rubric; magnitude stays null when the evidence is insufficient.
Unknown is not neutral

CHALLENGE 03 | UNRELIABLE INPUTS
Deduplication, freshness and quality gates prevent unsupported mock allocation changes.
No-op decisions and reasons remain inspectable



Provider failures are visible; historical replay is explicitly separate from live data.

MODULE A · TACTICAL MOCK INDEX

From interpreted signal to constrained allocation

1 **SIGNAL**

Company · sentiment · event · evidence

2 **ELIGIBILITY**

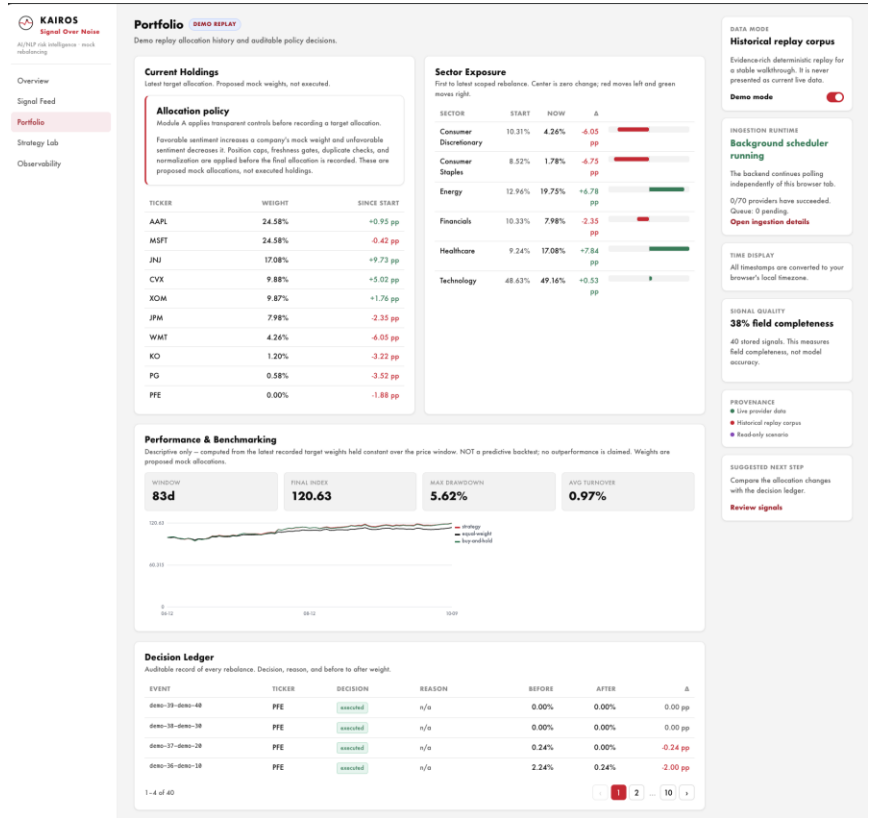
Freshness · quality · duplicate-consumption

3 **ALLOCATION RULE**

Favorable ↑ / unfavorable ↓; caps and non-negative weights

4 **AUDIT**

Renormalize to 100%; store before/after weights and reason



The decision ledger records proposed changes; no orders are executed.

REPRODUCIBLE EVALUATION

Three modes, three distinct claims

LIVE

Current public-provider inputs

Fresh inference; failures and skipped items are visible.

HISTORICAL REPLAY

Committed historical corpus

Repeatable end-to-end walkthrough through the real pipeline.

STRATEGY LAB

User-defined what-if scenario

Read-only counterfactual; does not alter live signals or ledger.

REPRODUCE LOCALLY

```
docker compose up --build
```

CPU-first · no API keys required by default · browser at localhost:5173

RESULTS, LIMITATIONS & NEXT STEPS

Implemented end-to-end; validation remains future work

DEMONSTRATED CAPABILITIES

- Live provider adapters, background ingestion and persisted raw items
- Company-specific signals with exact evidence and provenance
- Evidence-based deterministic impact; policy can abstain
- Module A mock rebalancing with before/after decision ledger
- Reproducible replay and read-only Strategy Lab scenarios

NOT YET VALIDATED

No held-out accuracy figures reported
No proven investment returns
No production-scale throughput benchmark

NEXT: MEASURE BEFORE SCALING

Evaluate labeled NLP fixtures; benchmark latency, queue depth and throughput before changing workers or storage.

Result: a traceable source → signal → policy → mock decision workflow, not a trading system.